

DSA 210 — Final Report (Spring 2025–2026)

The Pulse of WallStreetBets: Analyzing the Impact of Social Media Discussion Volume on Stock Market Volatility and Trading Volume

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Dsa210-Introduction to Data Science

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Github: <https://github.com/beyriahmetturan-rgb/Dsa-210>

Abstract

This project studies whether activity on r/wallstreetbets (WSB) is associated with—and can help predict—next-day trading volume and volatility for three meme stocks (GME, AMC, BB) during 2020–2021. Reddit post data are enriched with daily market data from Yahoo Finance. The workflow includes cleaning, daily aggregation, exploratory analysis, formal hypothesis tests, and supervised machine learning with time-ordered cross-validation. Results show a strong statistical link between WSB mentions and next-day volume for all three tickers; links to next-day volatility are weaker and ticker-dependent. Machine learning models improve on a naive baseline for volume prediction in several settings, but performance is limited by a short daily sample and regime shifts during the meme-stock period. All conclusions are **correlational**, not causal.

1. Motivation

The 2021 meme-stock episode showed that retail attention on social media can coincide with extreme market activity. A practical question is whether **discussion volume on WallStreetBets** can act as an **indicator** of upcoming market behavior—especially **trading volume** and **volatility**—rather than only reflecting events after they occur.

Core Research Question: *"Do daily discussion volumes and attention spikes on r/wallstreetbets act as a leading indicator of upcoming trading volume and market volatility for meme stocks (GME, AMC, BB), or do they merely reflect events after they occur?"*

This project follows the DSA 210 data-science workflow: define questions, collect and enrich data, explore patterns, test hypotheses with appropriate statistics, and extend the analysis with machine learning while stating limitations clearly. The focus is on **daily** Reddit attention and **next-day** market outcomes for GME, AMC, and BB.

2. Data sources and preparation

2.1 Sources

Source	Description	Role
Reddit WSB posts (`reddit_wsb.csv`)	Kaggle dataset (G. Preda format): title, body, score, timestamps	Social-media signal
Yahoo Finance (`yfinance`)	Daily OHLCV for GME, AMC, BB	Enrichment (course requirement)

2.2 Pipeline (summary)

1. Convert `created` (Unix epoch) to UTC dates; fill missing timestamps where possible.
2. Combine `title` and `body`; count ticker mentions with regex (e.g. GME, \$GME).
3. Aggregate to one row per calendar day: posts, score_sum, score_mean, mentions_<TICKER>.
4. Merge with cached daily market files on date.
5. Define next-day targets: vol_next, absret_next, hl_next.

Sample: 122 trading days per ticker with valid next-day targets after the join.

Period: roughly September 2020 – August 2021 (limited by the Reddit CSV).

2.3 Reproducibility (repository)

Analysis is in `dsa210_analysis.ipynb` and can be rerun with:

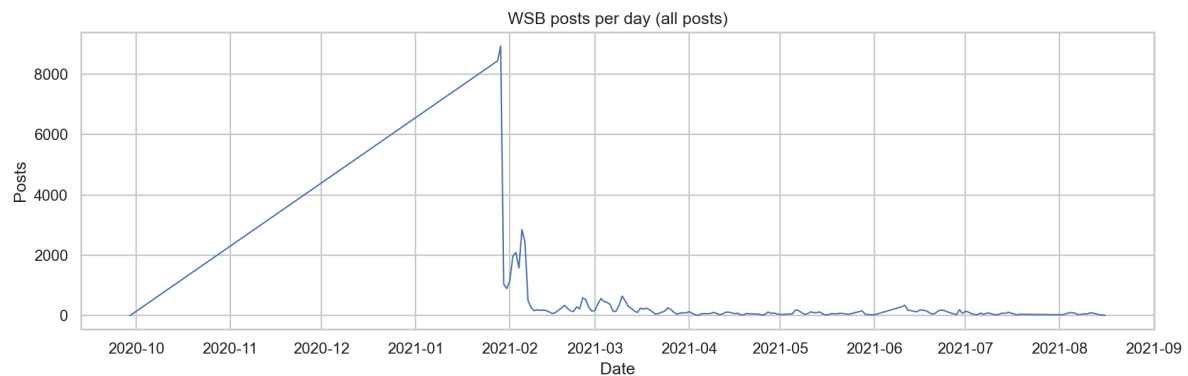
- `python -m src.run_eda_hypothesis --reddit_csv reddit_wsb.csv --tickers GME AMC BB`
- `python -m src.run_ml --from_tables outputs/tables --tickers GME AMC BB`

3. Exploratory data analysis

3.1 WSB activity over time

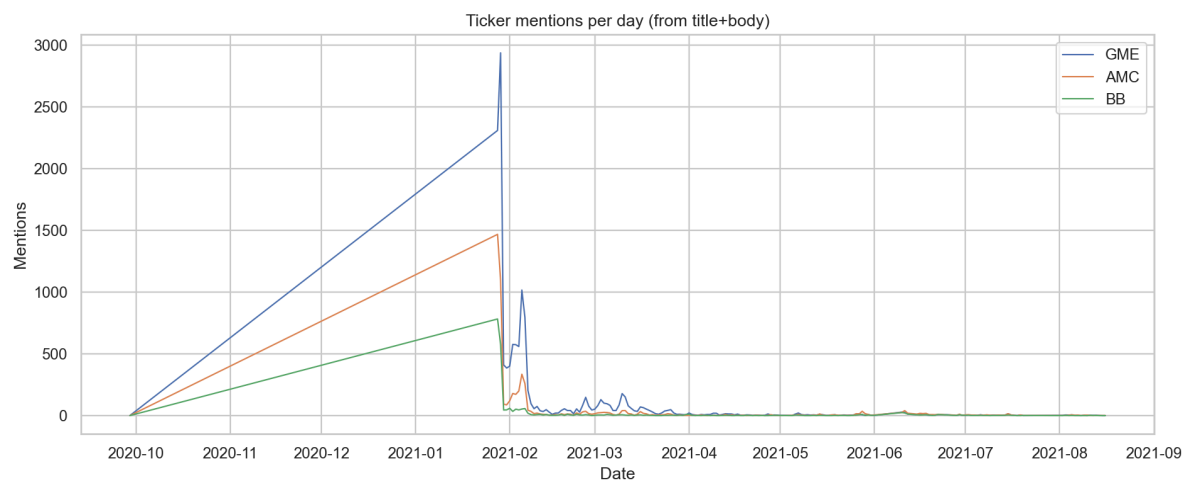
Posting activity is low in late 2020, then spikes sharply around the January 2021 meme-stock window.

Figure 1. Daily count of all WSB posts in the dataset



Ticker mentions follow a similar pattern; GME dominates attention during the squeeze weeks.

Figure 2. Daily mention counts for GME, AMC, and BB (regex on title+body)



3.2 Mentions vs next-day market variables

Scatter plots with regression lines suggest a **positive association** between same-day mentions and **next-day volume**, especially for GME. The link to next-day absolute return is noisier.

Figure 3. GME: daily mentions vs next-day trading volume.

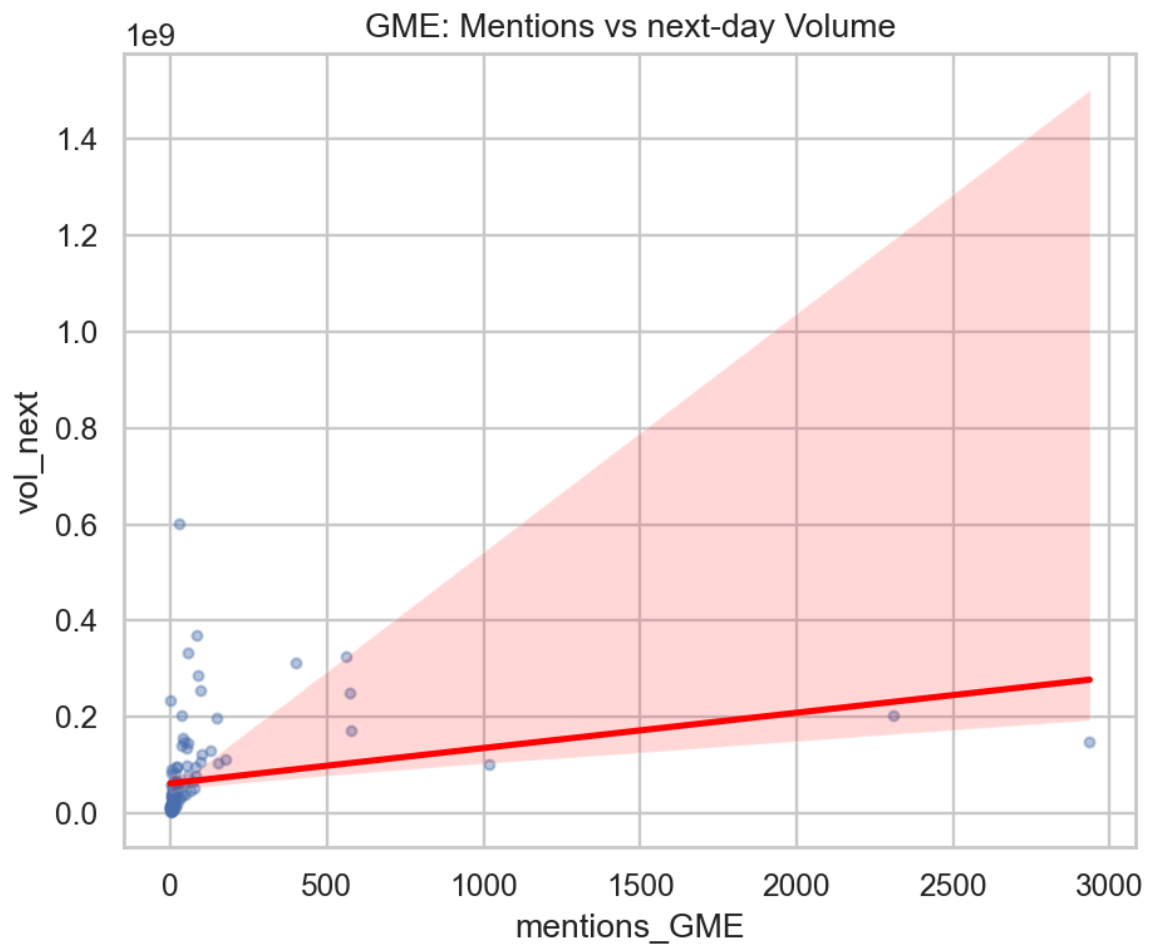
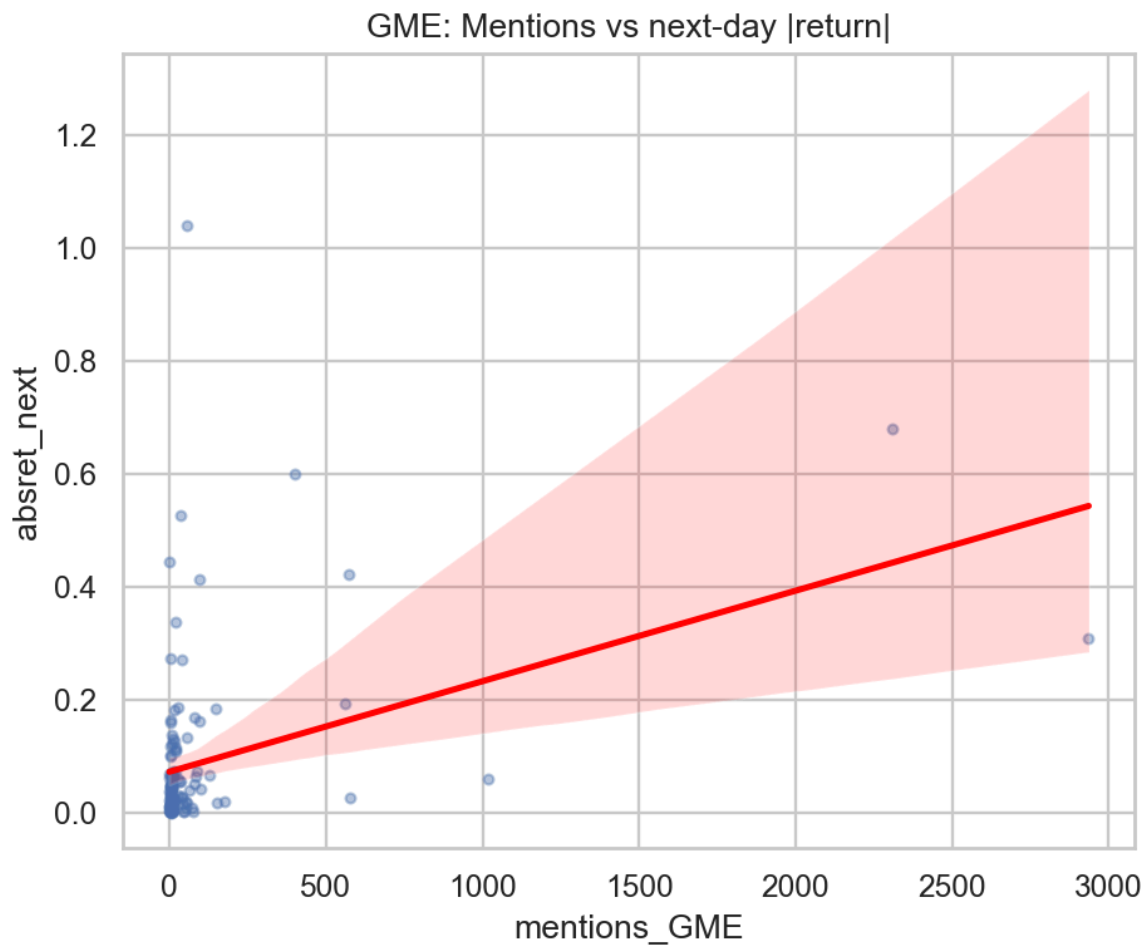


Figure 4. GME: daily mentions vs next-day absolute return.



Similar plots for AMC and BB are in the figures/ folder (AMC_*, BB_*).

EDA takeaway: Social attention and liquidity co-move in this sample; the strongest visual signal is for **volume**, not a clean volatility channel.

4. Statistical hypothesis tests

Design (per ticker):

- **High-attention days:** top quartile of daily mentions within that ticker's series.
- **H1 / H2:** Mann–Whitney U test—next-day volume (H1) and next-day absolute return (H2), high vs lower mention days.

- **H3 / H3b:** Spearman correlation between mentions and next-day volume (H3) or absolute return (H3b).

Results(from hypothesis_tests.csv in the repository):

Test	GME	AMC	BB
H1 — high mentions → higher next-day volume	Significant ($p \approx 7.3 \times 10^{-13}$; Cliff's $\delta \approx 0.87$)	Significant ($p \approx 8.6 \times 10^{-6}$; $\delta \approx 0.53$)	Significant ($p \approx 7.3 \times 10^{-11}$; $\delta \approx 0.76$)
H2 — high mentions → higher next-day \	Significant ($p \approx 0.026$; $\delta \approx 0.27$)	Significant ($p \approx 0.029$; $\delta \approx 0.26$)	Not significant ($p \approx 0.18$)
H3 — Spearman(mentions, next-day volume)	$\rho \approx 0.76$ ($p \approx 2.8 \times 10^{-24}$)	$\rho \approx 0.45$ ($p \approx 1.7 \times 10^{-7}$)	$\rho \approx 0.67$ ($p \approx 4.6 \times 10^{-17}$)
H3b — Spearman(mentions, next-day return)	$\rho \approx 0.32$ ($p \approx 2.7 \times 10^{-4}$)	Not significant ($p \approx 0.12$)	$\rho \approx 0.22$ ($p \approx 0.017$)

Interpretation: WSB attention is **most consistently** linked to **next-day trading volume**. Evidence for next-day volatility is **mixed**—significant for GME and AMC on Mann–Whitney (H2), but not for BB; Spearman on returns is weak for AMC.

5. Machine learning

5.1 Setup

- **Regression target:** $\log_{10}(\text{vol_next})$ from same-day Reddit features, same-day market proxies ($\log_{10}(\text{volume})$, abs_return , hl_range), and one-day lags.

- **Classification:** predict whether next-day volume is “high” ($\geq$ median on the **training fold** only).

- **Models:** Ridge, Random Forest, Gradient Boosting (regression); logistic regression and Random Forest (classification).
- **Validation:** TimeSeriesSplit (5 folds)—no random shuffle of trading days.

5.2 Regression (MAE vs naive mean forecaster)

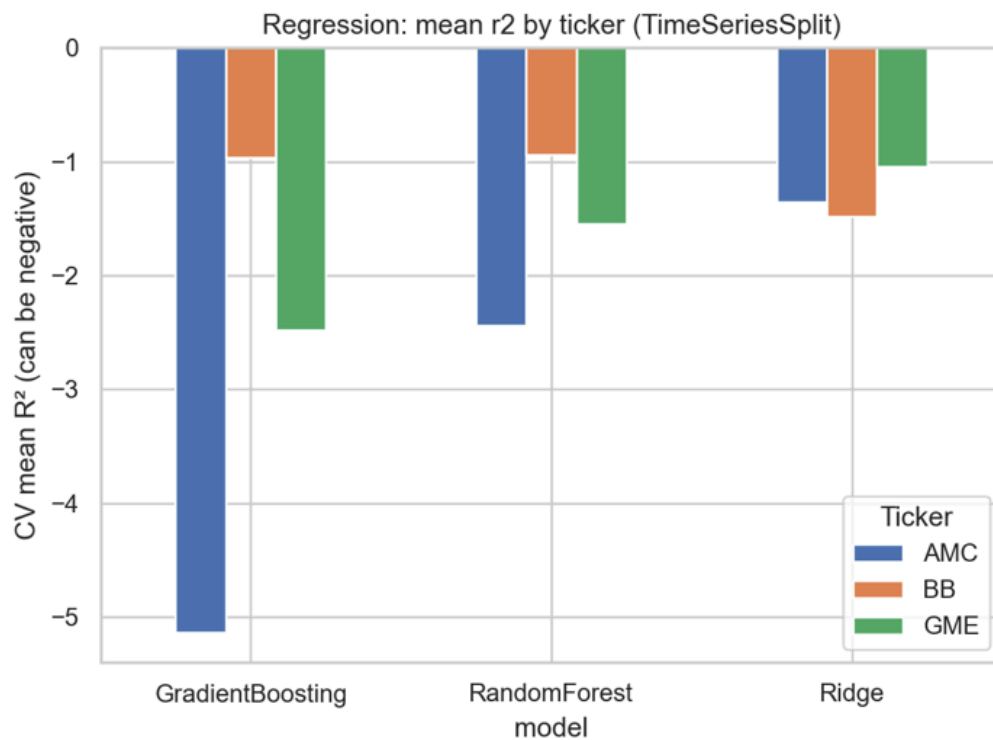
Ticker	Best Model (MAE)	MAE improvement vs Naive(%)
GME	Ridge	≈ 43%
AMC	Ridge	≈ 20%
BB	Random Forest	≈ 41%

R^2 on held-out folds is often **negative** during volatile windows even when MAE beats the naive baseline—**MAE improvement** is the more honest metric here.

Figure 5. Cross-validated mean MAE by model and ticker.



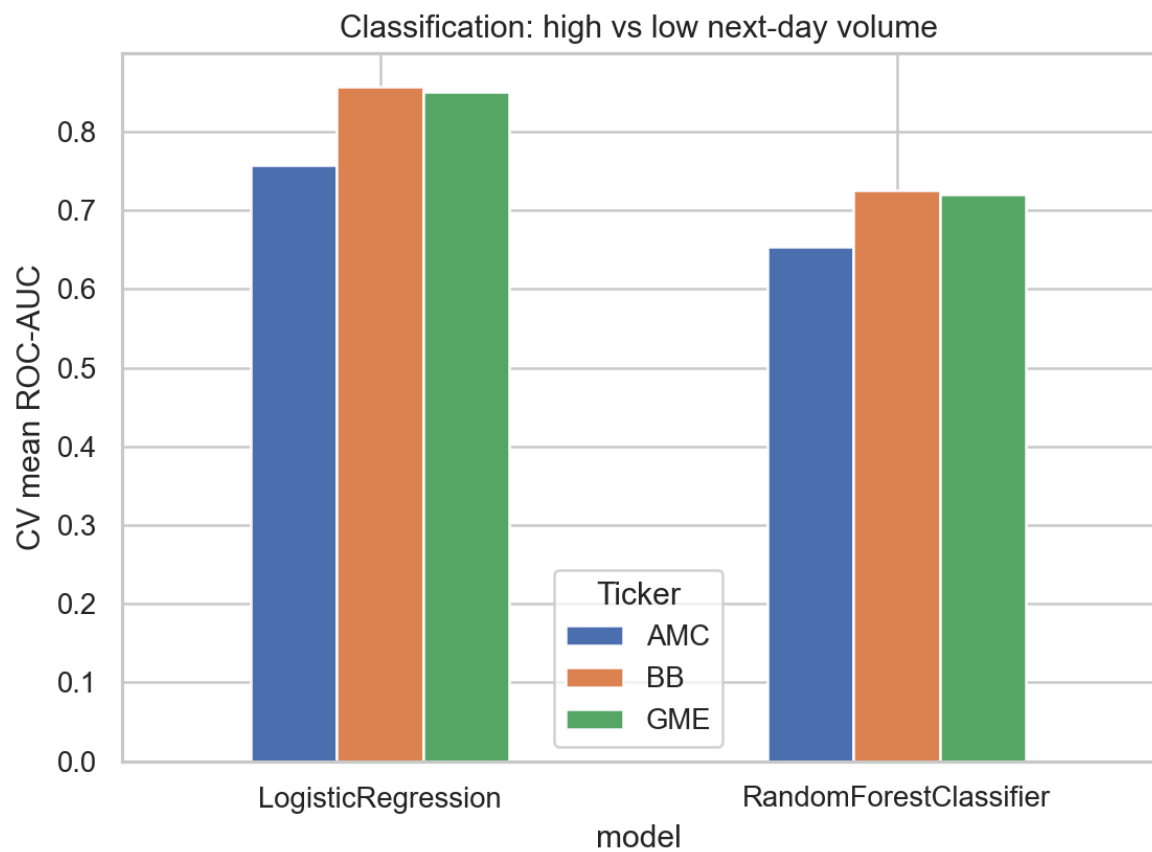
Figure 6. Cross-validated mean R^2 (often negative under regime shift)



5.3 Classification (ROC-AUC)

Ticker	ROC-AUC(Logistic)	ROC-AUC(RF)	Comment
GME	≈ 0.85	≈ 0.72	High accuracy (~ 0.89) but low F1 (~ 0.23)—imbalanced “high volume” class
AMC	≈ 0.76	≈ 0.65	More balanced F1 (~ 0.50 logistic)
BB	≈ 0.86	≈ 0.73	RF: moderate F1 (~ 0.53)

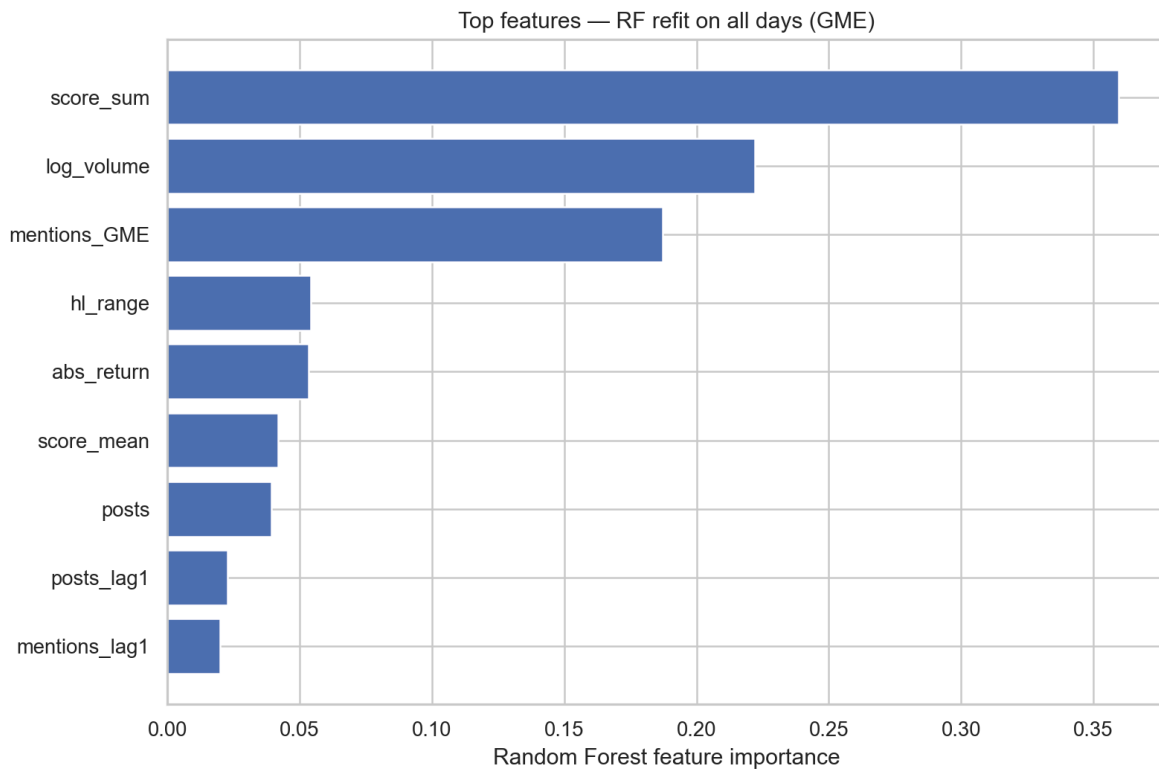
Figure 7. Cross-validated ROC-AUC for high vs low next-day volume



5.4 Feature importance (GME, descriptive)

Random Forest refit on all days (not for trading): **score_sum**, **log_volume**, and **mentions_GME** are among the top features.

Figure 8. Top-10 Random Forest feature importances for GME.



6. Integrated findings

- 1. Proposal delivered:** Reddit + market enrichment, EDA, hypothesis tests, and ML on GME, AMC, BB.
- 2. Volume:** Strong evidence that higher WSB mention days are followed by **higher next-day volume** (all three tickers).
- 3. Volatility:** Weaker and **ticker-dependent** ;BB shows no significant H2 group difference.
- 4. ML:** Models can **beat a naive mean forecast** for log-volume (especially GME and BB), but **R² is unreliable** under time-series CV in this episode.
- 5. Causality:** We **do not** claim Reddit posts **cause** market moves—only that they **co-move** in this historical window.

7. Personal observations

- 1. Regime effects:** Most signals sit in early 2021; models trained on “quiet” days fail on squeeze weeks—this explains negative R^2 with better MAE than naive.
- 2. Volume vs volatility:** WSB works better as a liquidity / attention indicator than a clean **volatility** indicator at daily frequency.
- 3. Mention extraction:** Regex counts can over-count tickers used as words; NLP entity linking could change effect sizes.
- 4. Same-day market state:** Same-day volume is very predictive; mentions add information but do not replace market features.
- 5. Metrics:** For this project, regression MAE tells a clearer story than classification accuracy under class imbalance.

8. Limitations and Future Work

8.1 Data Limitations

- **Intraday Timing and Daily Aggregation:** Compiling all Reddit posts into a single calendar day flattens the temporal dynamics of social media. In real financial markets, a viral post at 10:00 AM might trigger a massive trading volume surge at 11:30 AM. By aggregating data daily, this intraday lead-lag relationship is completely hidden, making it impossible to capture immediate market reactions.
- **Regex-based Mention Extraction:** Utilizing raw regular expressions (e.g., matching "GME" or "\$GME") only tracks the discussion volume but completely ignores contextual nuance and text sentiment. This approach introduces two issues: it allows for false positives (e.g., the ticker "BB" can easily be confused with normal English text or other abbreviations), and it treats a highly pessimistic post ("Short GME, it's over") identically to an optimistic one ("GME to the moon").
- **Short Sample and Single Historical Regime:** The combined dataset yields a final sample of only 122 trading days per ticker, spanning roughly from September 2020 to August 2021. More importantly, this specific period represents a highly anomalous, black-swan market regime driven by unprecedented retail coordination (the 2021 meme-stock craze). Consequently, the predictive signals found in this dataset are heavily regime-dependent and are highly unlikely to generalize to standard, less volatile market conditions.

8.2 Methodological and Machine Learning Limitations

- **Strictly Correlational Design:** This project establishes association, not causation. While the Mann-Whitney U and Spearman correlation tests confirm that high social media attention co-moves with next-day trading liquidity, we cannot methodologically prove whether Reddit hype *causes* market volume, whether price actions cause users to post more, or if a confounding third variable (such as mainstream media coverage) drives both simultaneously.

- **Regime Shifts and Negative R^2 in Validation:** Because the dataset transitions rapidly from a quiet regime to an extreme squeeze window, training models chronologically via `TimeSeriesSplit` creates a major mismatch. Models trained entirely on "quiet" historical blocks fail catastrophically when evaluated on subsequent high-volatility weeks. This structural regime shift explains why regression models often yield a negative R^2 on held-out folds, even when their Mean Absolute Error (MAE) successfully outperforms a naive baseline.
- **Class Imbalance in Liquidity Prediction:** Reframing next-day volume into a binary classification task ("high" vs "low") suffers from extreme class imbalance during non-squeeze weeks. This skew flattens the classification metrics; models can achieve a deceptively high overall accuracy (~89%) by simply dominant-class guessing, while their F1-score stays low (~0.23) because they fail to capture rare, extreme liquidity spikes accurately.

8.3 Future Work

- **Transition to Intraday Granularity:** Future extensions should utilize high-frequency market bars (e.g., 5-minute or 1-hour intervals) aligned precisely with UTC timestamps of Reddit submissions. This would allow for a proper event-study or Granger-causality framework to observe the exact microsecond flow of information from social boards to order books.
- **Advanced NLP and Sentiment Modeling:** Instead of simple keyword regex counting, future iterations should implement advanced Natural Language Processing (NLP) pipelines—such as FinBERT or VADER—specifically tuned for financial slang. This would convert raw mention frequencies into directional sentiment scores (Net Bullishness), filtering out irrelevant noise and hostile short-selling discussions.
- **Incorporation of Control Tickers:** To verify whether these social media dynamics are unique to speculative meme stocks, the analysis should be expanded to include a control group of traditional, institutional stocks (e.g., Apple, Microsoft) during the same time frame to evaluate if retail buzz exhibits the same predictive power across broader asset classes.

9. Academic Integrity and AI Usage Disclosure

This report and analysis are my own original work for DSA 210. In accordance with the course academic integrity policy, I am fully disclosing the use of AI tools in this project.

I used Cursor (an AI-powered IDE) as my primary development environment. Consequently, AI assistance was integrated into my workflow. The specific areas of AI assistance were:

- **Syntax and Boilerplate Code:** Cursor's autocomplete feature was used to generate standard pandas data manipulation syntax and matplotlib/seaborn plotting structures faster.
- **Debugging:** When encountering `KeyError` or index misalignment issues during the joining of Reddit and Yahoo Finance datasets, Cursor was utilized to identify the problematic lines and suggest syntax fixes.

- Documentation Formatting: AI was used to format the markdown tables and structure the text in the README and this final report.

Important Note: All critical analytical decisions—including the formulation of the research questions, the choice of non-parametric tests (Mann-Whitney and Spearman), the selection of TimeSeriesSplit to prevent data leakage, and the interpretation of the financial/social metrics—were made entirely by me. The AI was strictly a coding assistant, not an analyst. All numbers and conclusions were manually verified and run on my local environment.

References

- G. Preda, Reddit WSB posts dataset (Kaggle), `reddit\_wsb.csv`.
- Yahoo Finance via the `yfinance` Python package.
- DSA 210 course materials, Sabancı University, Spring 2025–2026.